

Skills checks cover pre-requisite material needed for the second unit of the course, derivatives. Answer the following questions neatly and completely. Calculators are not allowed.

1. Express the function $y = \left(\sqrt[3]{x+6}\right)^2 + 2$ as a composition of functions $f(g(x))$.

2. Express the function $y = \frac{1}{2} \cos(x^3 + 2)$ as a composition of functions $f(g(x))$.

3. Express the function $y = \sqrt{e^x - 2}$ as a composition of functions $f(g(x))$.

4. Solve the following equation for y : $4x^2 - 3x + xy = 2y + 5x^2y$.

5. Solve the following equation for f : $\frac{1}{f} + \frac{1}{g} = \frac{1}{h}$.

6. Solve the following equation for y : $\cos(x)(y+1) = 5y - 3$.

7. Find the equation of the line that contains the point $(2, 4)$ and is parallel to the line $2y = 7x - 4$.

8. A function $f(x)$ is linear and passes through the point $(-3, 1)$. If the slope of the line is 2, find the value of $f(6)$.

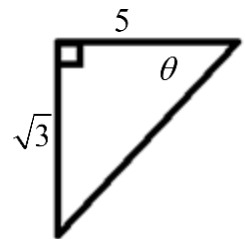
9. Find the equation of the linear function that contains the point $(2, 3)$ and is perpendicular to the line $y + 7 = 3x$.

10. Expand the following using properties of logarithms: $\ln\left(\frac{\sqrt{x}}{x^5(x-2)}\right)$.

11. Expand the following using properties of logarithms: $\ln\left(\sqrt[3]{\frac{x+3}{x-5}}\right)$.

12. Expand the following using properties of logarithms: $\ln\left(\sqrt{x^2(x+1)(x+2)}\right)$.

13. Create a trig. ratio that corresponds to the triangle.



14. Compute the exact value of $\tan\left(\frac{4\pi}{3}\right)$.

15. Create a trig. ratio that corresponds to the triangle.

